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Reinforcement Learning of Adaptive Virtual Damping in Variable Admittance Control for Tight Clearance Peg-in-Hole Assembly

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***Abstract*—Tight clearance peg-in-hole assembly requires precise alignment and compliant contact regulation, which challenges conventional fixed parameter controllers and reinforcement learning methods with low sample efficiency. This paper proposes a reinforcement learning-based variable admittance framework for robotic assembly, where an outer loop admittance controller ensures safe interaction and virtual damping parameters are adaptively adjusted. Learning efficiency and stability are enhanced via prioritized experience replay, a two-layer fuzzy reward capturing insertion efficiency and interaction safety, and a clearance-oriented curriculum strategy. The resulting method is termed Twin Delayed Deep Deterministic Policy Gradient with prioritized experience replay and clearance oriented curriculum Learning (TD3-PER-CCL). Experiments on a real robotic platform demonstrate higher success rates, faster convergence, and more stable force/torque responses compared with baseline methods.**

***Index Terms*—Peg-in-hole Assembly, Reinforcement Learning, Curriculum Learning, Admittance Control**

I. Introduction

Peg-in-hole assembly, as one of the most representative assembly operations in discrete manufacturing, is widely encountered in production processes spanning aerospace, automotive, and precision electronics manufacturing [1]–[3]. Existing methods for peg-in-hole assembly can be broadly grouped into three categories. First, vision based or geometry based approaches estimate the relative pose and guide the alignment and insertion [4]. While such methods can provide global guidance, their performance may be sensitive to environmental factors and may become less reliable in complex or partially observed settings. Second, model based compliance control, such as impedance/admittance control, has been widely adopted to regulate interaction forces and protect both the robot and the workpiece [5]. However, the performance of compliance control depends heavily on properly selected virtual parameters, which are typically tuned heuristically and may not adapt well to varying contact conditions and task difficulties. Third, learning based approaches, particularly deep reinforcement learning (DRL), have been increasingly investigated to reduce reliance on explicit contact modeling and to enable robots to acquire assembly skills through interaction. Nevertheless, DRL for tight clearance assembly often suffers from sample inefficiency and unsafe exploration, and it is also nontrivial to design a reward function that simultaneously encourages fast insertion and safe force regulation [6].

To address the above issues, this paper proposes TD3-PER-CCL, a reinforcement learning–based variable admittance assembly framework that integrates TD3 with prioritized experience replay, fuzzy reward shaping, and clearance-oriented curriculum learning along the clearance dimension. The key idea is to integrate an outer loop admittance controller as a safety critical compliance layer for contact interaction, while using DRL to adapt the virtual damping coefficient B, which is the most influential admittance parameter, thereby enhancing insertion efficiency and robustness. In each control cycle, the admittance controller converts force/torque feedback into incremental motion corrections, and the TD3-PER agent outputs the damping parameter to modulate the compliance behavior under different contact situations. In addition, a fuzzy reward mechanism is designed to encode multiple objectives without requiring an exact analytical reward model, thereby reducing the risk of misleading reward signals and local optima. Furthermore, a curriculum learning strategy is introduced by progressively tightening the hole clearance during training, enabling the agent to first master stable insertion under easier clearances and then transfer the acquired skills to more challenging, tight clearance conditions.

The main contributions of this paper are summarized as follows:

- A TD3-based variable admittance control framework is developed for tight clearance peg-in-hole assembly. Within this framework, a TD3 agent with prioritized experience replay is employed to learn the optimal virtual damping parameter B. In addition, a two-layer fuzzy reward mechanism is designed to simultaneously capture

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insertion efficiency and interaction safety in rich contact scenarios.
- A clearance oriented curriculum learning strategy is introduced to alleviate training instability under limited data. The curriculum schedules tasks along the clearance dimension by gradually tightening the peg-hole clearance.

II. RELATED WORK

Robotic peg-in-hole assembly has long been a core topic in industrial robotics. Assembly operations account for a substantial share of total production cost. Small clearance conditions require both high positioning accuracy and high compliance at the same time. Prior studies generally fall into three groups: model based force and position strategies, compliance control with carefully tuned impedance or admittance parameters, and learning based methods that reduce reliance on explicit contact modeling. Despite this progress, learning based approaches still face challenges in sample efficiency, safe exploration, and reward design.

A classical line of research estimates misalignment and contact state from force/torque (F/T) feedback, and then generates corrective motions using analytic rules or contact models. Liao et al. explicitly incorporate friction into peg-in-hole assembly by estimating a friction coefficient and deriving a friction-aware mapping between the measured contact wrench and the misalignment angle [7]. In addition to friction-aware estimation, some works emphasize posture correction as a prerequisite stage before insertion. For instance, Liu et al. propose a two stage posture estimation and adjustment pipeline for precise peg-in-hole: a rough adjustment using an MLP classifier trained with force–angle samples, followed by a fine adjustment based on a fuzzy finite variable admittance controller [8]. While these model centric approaches can be effective, their performance often depends on the correctness of modeling assumptions and on carefully designed decision logic, which may not transfer smoothly across varying clearances and contact conditions.

To guarantee safety in rich contact insertion, impedance and admittance control is widely used as an underlying compliance layer. However, the effectiveness of compliance control hinges on appropriate virtual parameters, and manual tuning can be brittle under changing contact situations. A representative direction is to use RL to output controller parameters rather than direct end effector motions. Li et al. propose a DDPG based variable parameter admittance control method integrated with a fuzzy reward mechanism, aiming to avoid directly applying RL actions to low level control and to reduce the need for precise analytical reward models [9]. Similarly, Chen et al. introduce a combined RL framework to train a variable impedance controller using two DQNs [10]. Liu et al. proposed a model based meta-reinforcement learning approach for industrial insertion assembly tasks, enabling rapid online adaptation by updating the dynamics model in real time through interaction with the environment [11]. Zhang et al. employ TD3 to optimize admittance parameters and design nonlinear reward shaping to accelerate training and avoid local optima [12].

Despite the above progress, DRL based assembly often suffers from large data requirements and unsafe exploration. A common remedy is to keep an underlying controller and enhance learning efficiency with replay prioritization and curriculum strategies. Peng et al. propose combining a PD force controller with TD3, they also introduce curriculum learning to improve training stability and adaptability across peg-in-hole tasks [13]. Liu et al. advocate multi phase learning that integrates human knowledge, model based components, and model free RL to improve learning efficiency. [14]. However, tight clearance insertion still faces practical challenges in sample efficiency, safe exploration, and reward shaping. To address these issues, this paper proposes TD3-PER-CCL. It is a TD3-PER driven variable admittance framework that learns adaptive virtual damping for rich contact assembly. This paper introduces a fuzzy reward mechanism to encode insertion efficiency and interaction safety and adopts clearance oriented curriculum learning. The curriculum progressively reduces the clearance to stabilize training and improve generalization across varying clearance conditions.

III. METHOD

This section presents TD3-PER-CCL, a variable admittance reinforcement learning framework for tight clearance peg-in-hole assembly. Fig. 1 shows the overall architecture. The framework contains three parts: a variable admittance controller, a TD3 agent with prioritized experience replay, and a clearance oriented curriculum learning scheduler. A two-layer fuzzy reward is used to shape the learning signal.

A. Variable Admittance Control

To ensure compliant contact interaction, this paper employs an outer loop admittance controller that maps the measured force/torque wrench to incremental motion commands. The continuous time admittance model is given by:

$$M\,\Delta\ddot{X} + B\,\Delta\dot{X} + K\,\Delta X = \Delta F \tag{1}$$

where ΔX denotes the admittance induced motion correction, and F is the measured external wrench (force/torque). M, B, and K are the virtual inertia, damping, and stiffness matrices, respectively. Moreover, $\Delta F = F_d - F$ represents the deviation between the desired contact wrench F_d and the wrench measured by the six axis force/torque sensor. A key challenge in admittance control is to identify the parameters M, B, and K in the admittance model. From Eq. (1), the transfer function of this control model is given as follows:

$$H(s) = \frac{\Delta X(s)}{\Delta F(s)} = \frac{1}{Ms^2 + Bs + K}$$

Since fine insertion mainly involves axial motion and tilt correction, this paper focuses on three controlled components,

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i.e., z, r_x, and r_y. Accordingly, the admittance parameters are structured as:

$$\begin{cases} M = [M_z,\, M_{r_x},\, M_{r_y}] \\ B = [B_z,\, B_{r_x},\, B_{r_y}] \\ K = [K_z,\, K_{r_x},\, K_{r_y}] \end{cases}$$

In our framework, M and K are kept constant, while the RL agent adapts B. In tight clearance insertion, the damping parameter has the most direct influence on contact stabilization, oscillation suppression, and compliant force regulation. M and K are kept constant to ensure the physical stability of the assembly task. Simultaneous online adaptation of M, B, and K would enlarge the action space and increase the difficulty of policy learning, which may reduce training stability and lead to unsafe exploration in real robot experiments. Therefore, to achieve a practical balance between adaptability and safety, this work only optimizes B online, while M and K are fixed to stable empirically tuned values.

B. TD3-PER for Adaptive Damping Learning

The tight clearance peg-in-hole assembly task is formulated as a continuous Markov decision process, denoted by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, where $\mathcal{S}$ and $\mathcal{A}$ are the state and action spaces, respectively, $\mathcal{P}$ represents the unknown transition dynamics of the contact environment, $\mathcal{R}$ is the reward function, and γ is the discount factor. To capture the complex contact conditions and relative misalignment during insertion, the state vector s_t fuses multimodal proprioceptive and haptic feedback. It is defined as an 8-dimensional vector:

$$s_t = [F_x,\, F_y,\, F_z,\, M_x,\, M_y,\, \theta_x,\, \theta_y,\, h]$$

where F_i and M_i denote the contact forces and torques measured by the wrist force/torque sensor, which serve as critical indicators of jamming or collision. θ_x and θ_y represent the angular posture deviations of the peg relative to the hole axis, and h indicates the current vertical insertion depth. This composite state allows the agent to perceive both geometric alignment and contact interaction dynamics.

Unlike conventional kinematic control policies that directly output position increments, our framework adopts a variable admittance policy to enhance interaction safety. Specifically, the action a_t modulates the virtual damping parameters in the admittance model. The action vector is defined as:

$$a_t = \left[B_z,\, B_{r_x},\, B_{r_y}\right]$$

where B_z governs the damping behavior along the insertion axis, and B_{r_x} and B_{r_y} regulate the rotational compliance about the x and y axes, respectively. By adaptively tuning these parameters, the agent learns to optimize the trade-off between insertion efficiency and contact force regulation.

To learn a continuous damping policy $a_t = \pi_\phi(s_t)$, this paper adopts TD3 [15] with prioritized experience replay. Two critics, $Q_{\theta_1}(s,a)$ and $Q_{\theta_2}(s,a)$, are used to mitigate overestimation via clipped double Q, while delayed policy updates and target policy smoothing further improve training stability. Specifically, this paper maintains target networks $\bar{\pi}_\phi$, $\bar{Q}_{\theta_1}$, and $\bar{Q}_{\theta_2}$. Given a transition (s_t, a_t, r_t, s_{t+1}), the next action is generated by the target actor with clipped Gaussian noise:

$$\tilde{a}_{t+1} = \bar{\pi}_\phi(s_{t+1}) + \epsilon,\ \epsilon \sim \mathrm{clip}\big(\mathcal{N}(0, \sigma^2), -c, c\big)$$

The TD target is defined in Eq. (2), where the minimum of two target critics is adopted to alleviate overestimation:

$$y_t = r_t + \gamma \min_{i \in \{1,2\}} \bar{Q}_{\theta_i}(s_{t+1}, \tilde{a}_{t+1})\,. \tag{2}$$

Each critic is optimized by minimizing the Bellman error in Eq. (3):

$$\mathcal{L}(\theta_i) = \mathbb{E}\Big[\big(Q_{\theta_i}(s_t, a_t) - y_t\big)^2\Big],\ i \in \{1,2\} \tag{3}$$

To further improve stability, the actor is updated less frequently by maximizing the expected critic value:

$$\max_{\phi}\ \mathbb{E}[\,Q_{\theta_1}(s_t, \pi_\phi(s_t))]$$

Target networks are updated via Polyak averaging:

$$\bar{\theta}_i \leftarrow \tau\theta_i + (1-\tau)\bar{\theta}_i,\ \bar{\phi} \leftarrow \tau\phi + (1-\tau)\bar{\phi}$$

The transition sampling probability is computed according to Eq. (4), which biases training toward informative samples:

$$P(i) = \frac{p_i^\alpha}{\sum_k p_k^\alpha} \tag{4}$$

To reduce the induced bias, importance sampling weights are applied:

$$w_i = \left(\frac{1}{N\,P(i)}\right)^\beta$$

where N is the buffer size, and α and β control the prioritization strength and the degree of bias correction, respectively.

C. Fuzzy Reward Mechanism

Designing an analytic reward that simultaneously encourages fast insertion, low contact forces/torques, and stable alignment is nontrivial. Therefore, this paper introduces a fuzzy reward mechanism to encode multi objective signals and reduce sensitivity to modeling inaccuracies. The overall reward is defined as

$$r_t = r^{\mathrm{succ}} + r^{\mathrm{fuzzy}}$$

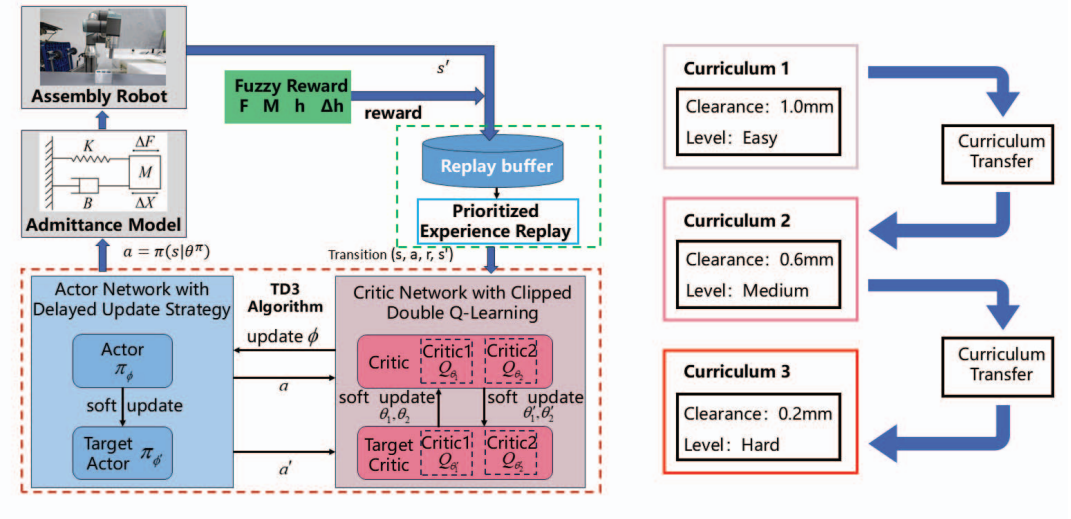
where r^{succ} is provided only when the robot successfully completes the assembly. Specifically, if the peg reaches the target position, the contact wrench does not exceed the prescribed safety limits, and the task is completed within the maximum number of steps, the episode is regarded as successful and the agent receives a positive reward. The success reward is defined as

$$r^{\mathrm{succ}} = k\left(1 - \frac{n}{n_{\max}}\right)$$

where n is the total number of steps in a single assembly episode, $n_{\max}$ is the predefined maximum number of steps, and k is the reward coefficient.

The term r^{fuzzy} denotes the negative reward assigned to each executed action and is obtained from the fuzzy reward

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(a) The framework of peg-in-hole assembly task (b) Curriculum transfer process

Fig. 1: The illustration of system framework and curriculum transfer process.

system. The penalty increases when the contact forces/torques are larger, the current insertion depth is shallower, or the per-step insertion increment Δh is smaller. The fuzzy system takes the current depth h, the incremental depth per step Δh, the maximum contact force $F_{\max}$, and the maximum contact moment $M_{\max}$ as inputs, where each input variable shares the same linguistic set $[VB,\ B,\ N,\ G,\ VG]$.

The fuzzy reward system evaluates the interaction state and generates a penalty based on the current insertion depth h, the per-step feed increment Δh, the maximum contact force among the three force components $F_{\max}$, and the maximum moment about the x and y axes $M_{\max}$. Using the maximum force and moment as inputs encourages compliant assembly behaviors.

A two-layer fuzzy reward structure is adopted to reduce computational complexity, as shown in Fig. 2. The system consists of three fuzzy sets and a total of 75 fuzzy rules. The first layer contains two fuzzy subsystems, namely the force moment subsystem and the current depth feed increment subsystem. The second layer takes the outputs of these two subsystems as inputs. Finally, the penalty value is obtained by defuzzification according to Eq. (5):

$$y(\mathbf{x}) = \frac{\sum_{r=1}^{25} c_r \prod_{k=1}^{d} \mu_{r,k}(x_k)}{\sum_{r=1}^{25} \prod_{k=1}^{d} \mu_{r,k}(x_k)} \tag{5}$$

where $\mathbf{x} = [x_1, x_2, \cdots, x_d]$ is the input vector and $y(\mathbf{x})$ is the defuzzified output. $\mu_{r,k}(x_k)$ denotes the triangular membership function associated with the k-th input variable x_k in the r-th rule, and different fuzzy subsets employ different triangular membership functions. c_r is the consequent representative output value of the r-th fuzzy rule, whose value is determined empirically.

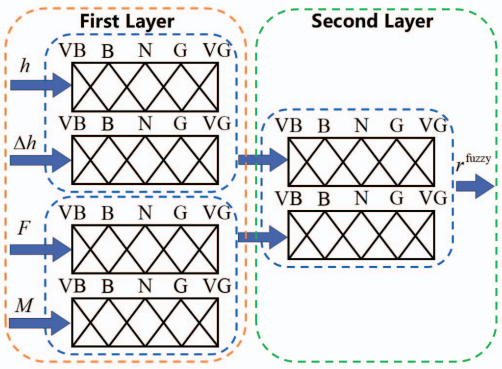


Fig. 2: The two-layer fuzzy reward system.

D. Clearance oriented Curriculum Learning

Curriculum Learning is a training strategy inspired by how humans acquire new skills. Its key idea is to organize training tasks in a progressively increasing order of difficulty, allowing the agent to first learn fundamental and easier skills and then gradually transition to more complex tasks on top of what has been mastered. This strategy has attracted substantial attention in deep reinforcement learning and has demonstrated strong performance in computer vision, robotic control, and skill learning tasks, significantly improving learning efficiency and generalization capability.

In peg-in-hole assembly policy training, a smaller clearance makes the task more challenging. Training directly under high difficulty conditions typically requires more training steps and incurs higher trial and error costs, leading to a substantial increase in training overhead. To address this issue, this paper

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introduces curriculum learning into the peg-in-hole policy training process and decompose the assembly task into three progressively harder curriculum levels, where the clearance size is used as the criterion of task difficulty. Specifically, a clearance of $1.0\,\mathrm{mm}$ corresponds to the easy level, $0.6\,\mathrm{mm}$ to the medium level, and $0.2\,\mathrm{mm}$ to the hard level. Training starts from Curriculum 1. When the number of training episodes exceeds 300 and the average success rate over the most recent 150 episodes in the current curriculum is above 80%, the curriculum automatically switches to the next difficulty level. The curriculum transition process is illustrated in Fig. 1b.

IV. EXPERIMENTS

This section introduces the experimental setup and reports comparative results for tight clearance peg-in-hole assembly.

A. Experimental platform

The assembly experiments were conducted on a UR3e robot platform with an on-robot six-dimensional force/torque sensor. The control computer was equipped with an Intel i7 CPU and an NVIDIA GeForce GTX 3070 GPU. The peg used in the insertion task had a diameter of 20 mm and a length of 50 mm. Peg-in-hole insertion tests were performed using holes with diameters of 20.2 mm, 20.6 mm, and 21.0 mm.

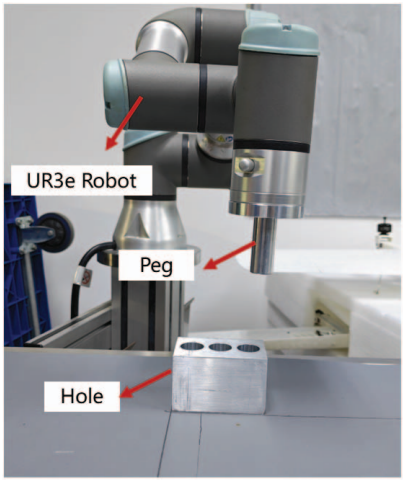


Fig. 3: The experimental platform of assembly.

B. Implementation Details

The TD3-PER agent adopted an actor-critic architecture, where the actor and each critic had two fully connected hidden layers with 256 neurons per layer and ReLU activations. The actor and critic learning rates were set to 1×10^{-4} and 1×10^{-3}, respectively, with $\gamma = 0.99$ and $\tau = 0.005$. For TD3, the target policy smoothing parameters were set to $\sigma = 0.2$ and $c = 0.5$. The replay buffer size was 1×10^{5}, and the batch size was 128. For PER, the prioritization exponent was set to $\alpha = 0.6$, while the importance-sampling exponent was linearly annealed from 0.4 to 1.0 during training.

The fixed admittance parameters were determined empirically through preliminary experiments. Specifically, $M = [1, 0.05, 0.05]$ and $K = [1000, 10, 10]$ for the z, r_x, and r_y directions, respectively, while the RL agent adjusted $B_z \in [1, 100]$, $B_{rx} \in [0.1, 10]$, and $B_{ry} \in [0.1, 10]$ online.

C. Results and Discussion

To validate the effectiveness of the proposed approach, three ablation settings were considered: TD3, TD3-PER, and TD3-PER-CCL. For fair comparison, all methods use the same state/action definitions and the same admittance based low level controller, only the learning components differ. Each method was trained for 1000 episodes, with the maximum number of time steps per episode set to 200. At the beginning of each episode, the peg was positioned above the hole with a randomized initial orientation perturbation to emulate assembly uncertainties. An episode was regarded as successful if the target insertion depth was reached within the step limit and the contact force/torque remained within prescribed safety constraints. After training, this paper additionally recorded the force/torque time series from a representative real insertion trial to further analyze interaction safety and compliance during the assembly process.

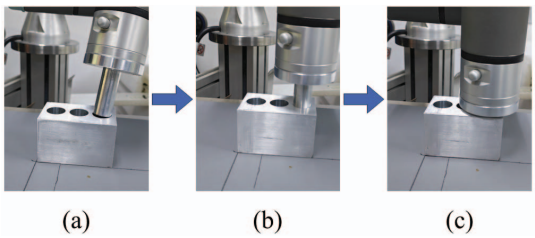


Fig. 4: The experimental process of assembly. (a) Initial contact. (b) Pose alignment. (c) Peg insertion.

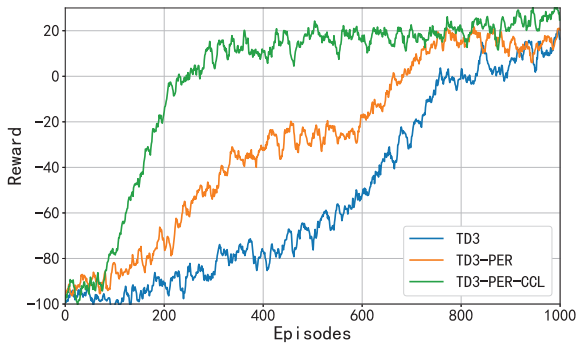


Fig. 5: Training reward curves of different algorithms.

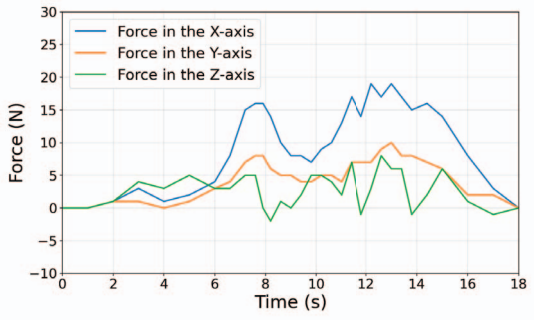
Fig. 5 shows that all methods start with low returns, reflecting ineffective contact alignment and insertion behaviors at the early exploration stage. However, clear differences emerge as training proceeds. TD3 exhibits the slowest improvement and larger fluctuations,TD3-PER improves both learning speed and stability by sampling high priority transitions more frequently. TD3-PER-CCL achieves positive returns significantly earlier and converges to a higher plateau.

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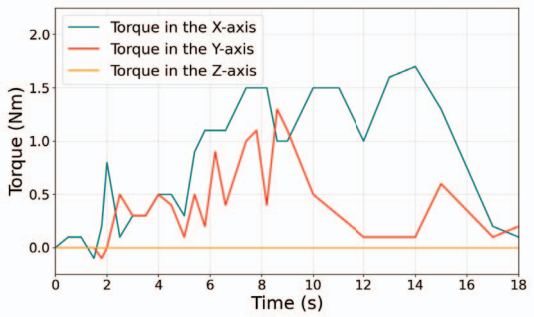
Table I reports the assembly success rate and the average number of steps. TD3-PER-CCL achieves the highest success rate and requires the fewest steps, outperforming TD3 and TD3-PER.

TABLE I: ASSEMBLY TRAJECTORIES FOR DIFFERENT ALGORITHMS

Method	Assembly Success Rate(%)	Average Assembly Steps
TD3	81.3	44
TD3-PER	85.8	37
ours	91.4	29



(a) The change curve of the force.



(b) The change curve of the torque.

Fig. 6: Changes in force and torque during assembly.

To further inspect interaction behavior, Fig. 6a and Fig. 6b plot representative force and torque trajectories of a successful insertion after training. The post training force trajectories exhibit a staged behavior. Forces rise when contact is first established, followed by transient lateral force peaks during the correction phase, and then decrease as the peg becomes well aligned and settles into the hole. The torque trajectories follow a similar pattern. Overall, the wrench profiles suggest compliant interaction without sustained large forces/torques, which is consistent with safe insertion behavior enabled by the admittance layer and adaptive damping policy.

V. CONCLUSION

This paper proposes TD3-PER-CCL, a reinforcement learning based variable admittance framework for tight clearance peg-in-hole assembly. The framework combines an outer loop admittance controller with a TD3 agent using prioritized experience replay to adapt the virtual damping parameter. It further introduces a two-layer fuzzy reward shaping scheme to capture both insertion efficiency and interaction safety. In addition, clearance-oriented curriculum learning is adopted to schedule training from large to tight clearances. Experiments show that TD3-PER-CCL improves assembly success rate, reduces assembly steps, and yields stable force/torque behaviors. CCL enables progressive skill transfer across different clearances and reduces reliance on manual parameter tuning. These properties support practical deployment in rich contact industrial assembly scenarios.In future work, we will further evaluate the generalization capability of the proposed framework in more challenging assembly scenarios, such as tasks involving non-cylindrical peg geometries and dynamic moving-target environments.

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